香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 24 Machine Learning: Model Selection

Model Selection

Question 1: Which supervised learning method should be selected, KNN or logistic regression?

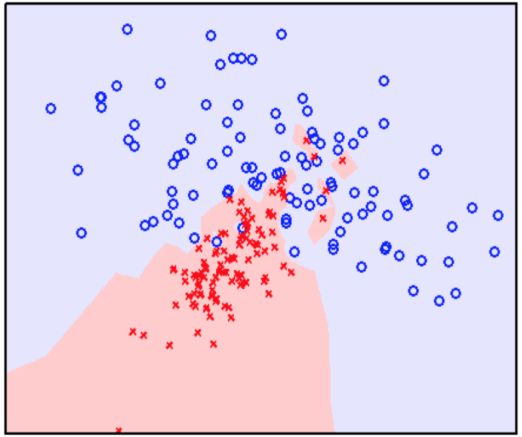
Question 2: How many neighbors (K) should be chosen in KNN methods?

Question 3: How to determine the best number of clusters for K-means?

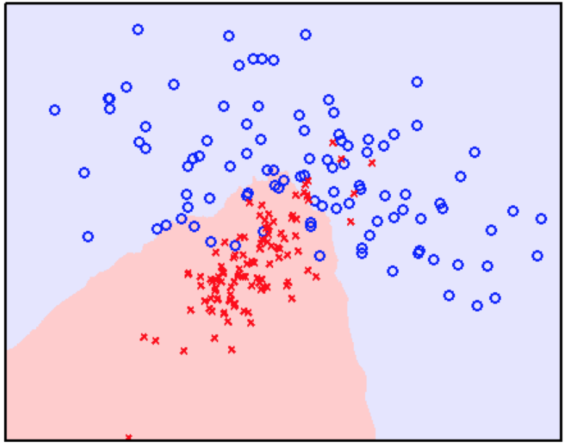
KNN

Overfitting

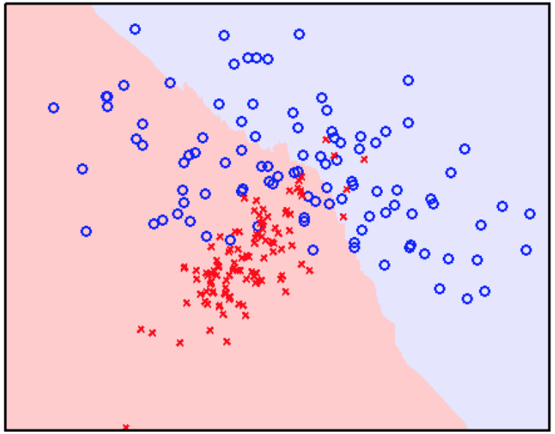
Underfitting



$K = 1$



$K = 25$



$K = 101$

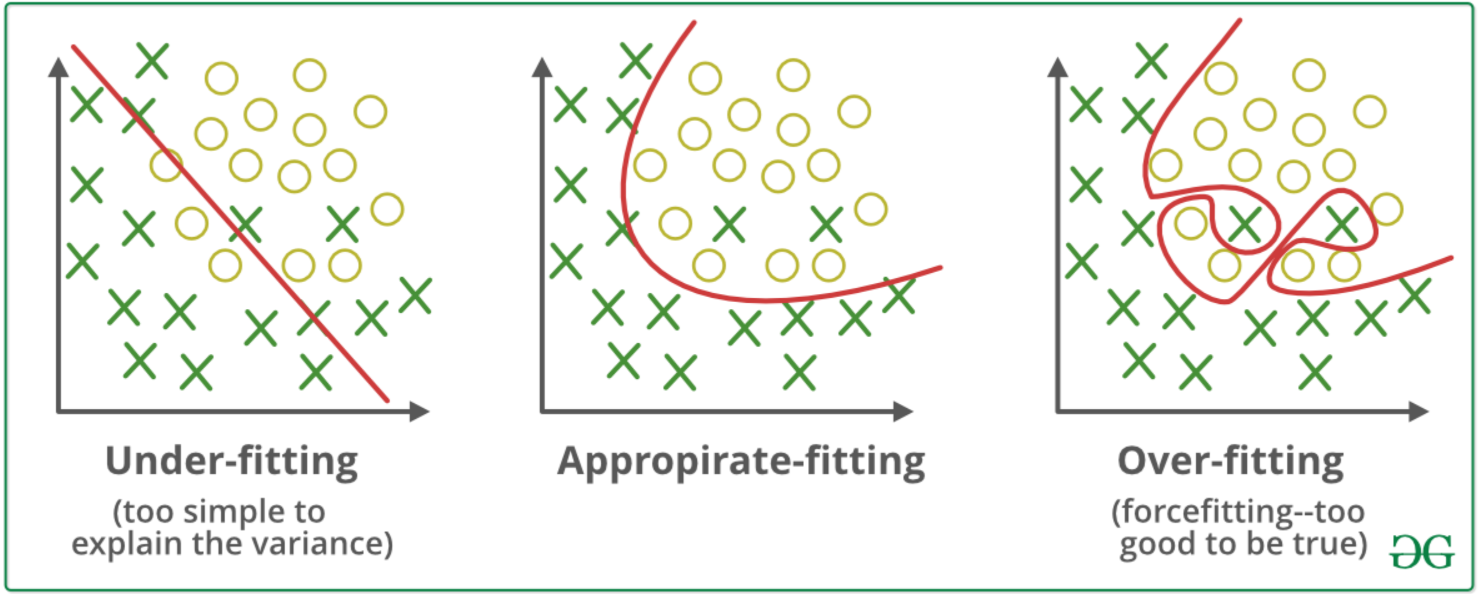
Overfitting and Underfitting

- Machine learning models are built to learn from training and test data, enabling them to make predictions on new, unseen datasets.

- A machine learning model is said to **overfit** the data when it learns patterns specific to the training data and make accurate predictions only on the training data (for testing data, prediction accuracy is low).

- A machine learning model is said to **underfit** when it fails to capture the key patterns or relationships between variables in both the training and test data.

Overfitting and Underfitting



Example: Nonlinear Regression Model

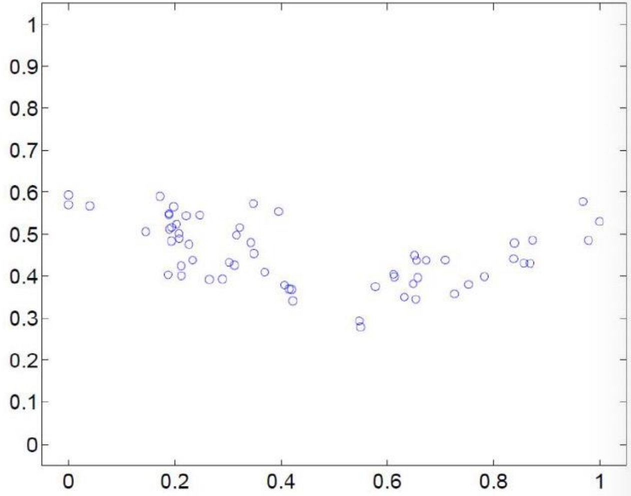
- Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \ldots + \theta_n x^n + \epsilon$$

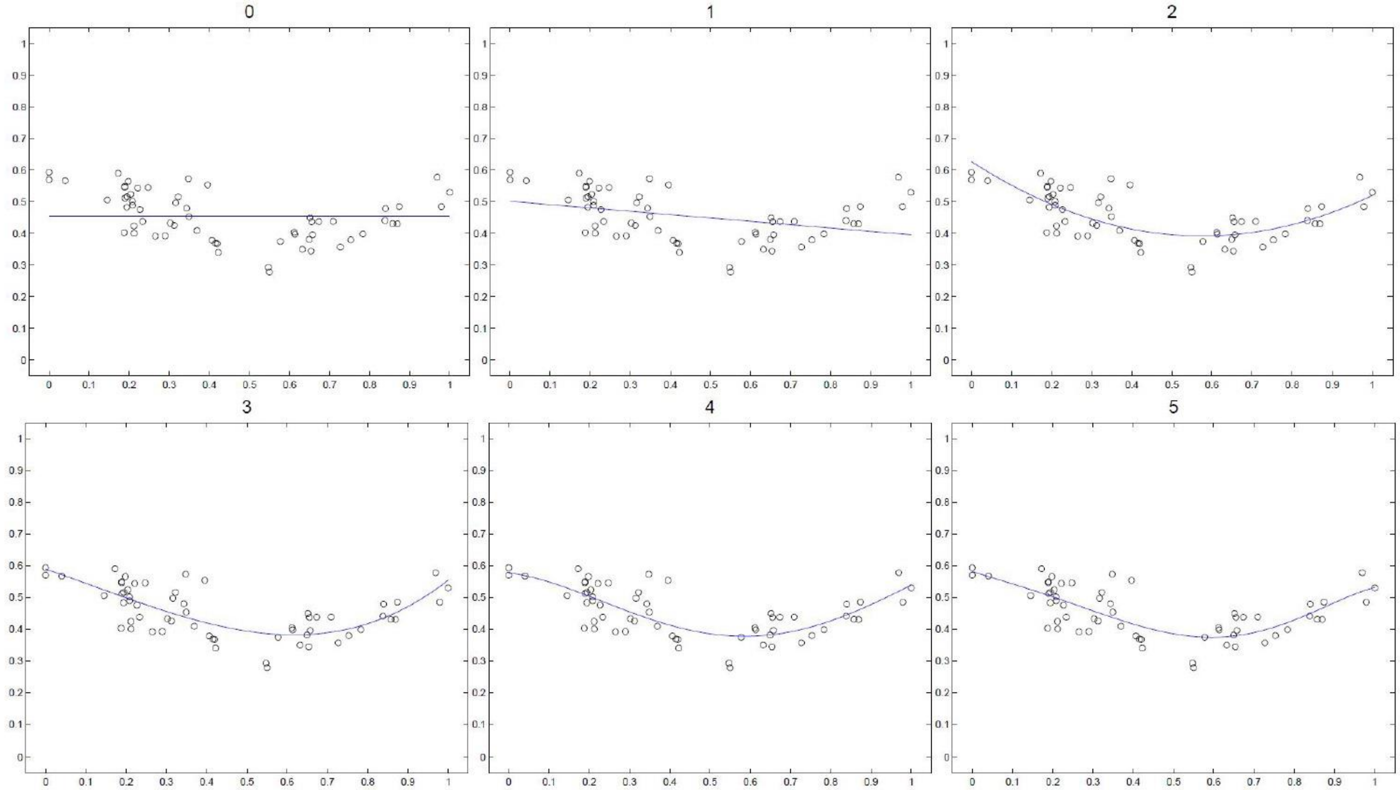
- Let $x = (1, x, x^2, \ldots, x^n)^\top$ and

$\theta = (\theta_0, \theta_1, \theta_2, \ldots, \theta_n)^\top$

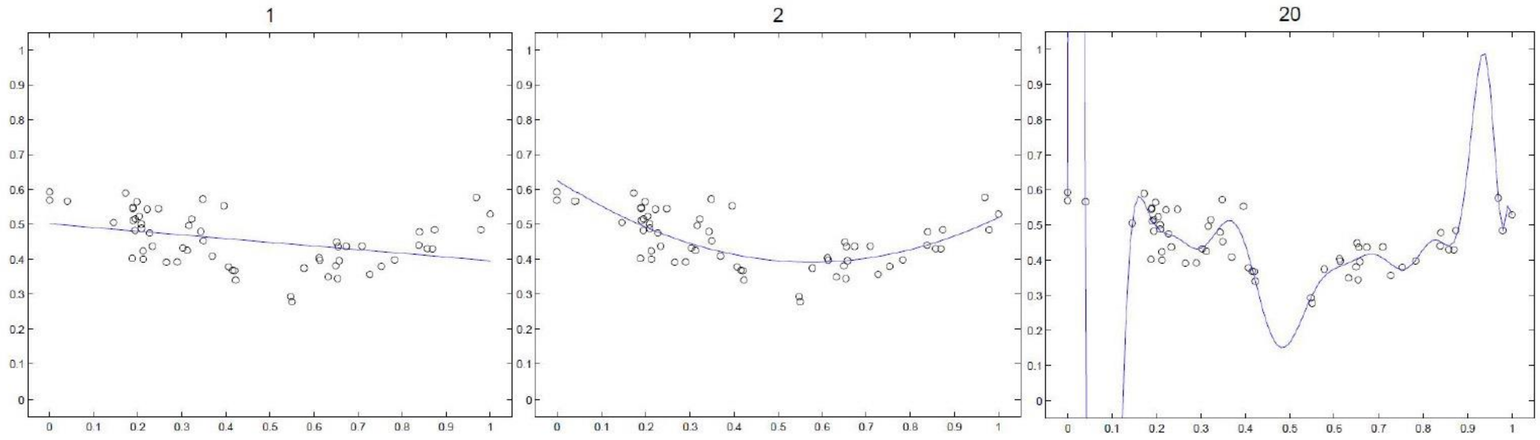
$$y = \theta^\top x$$



Increasing the maximal degree



Which one is better?



- Can we increase the maximal polynomial degree to very large, such that the curve passes through all training points?

- The optimization does not prevent us from doing that

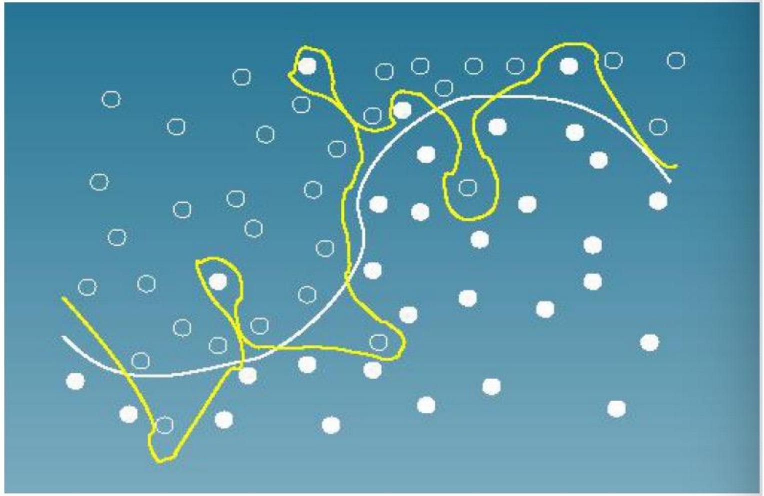
Example: Classification Model

- Logistic regression with polynomial features

$$p(y = 1 | x, \theta) = \frac{1}{1 + \exp\left(-\theta^\top x\right)}$$

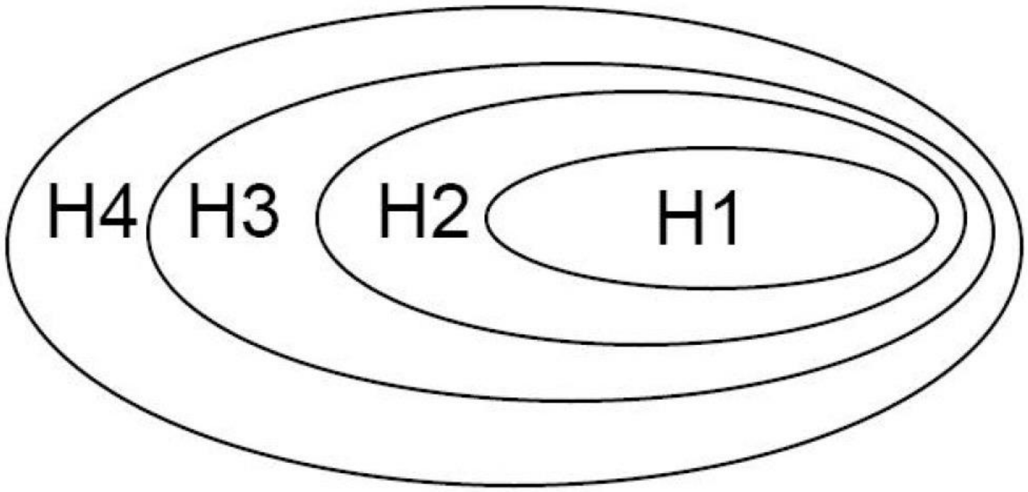
- Let $x = (1, x, x^2, \ldots, x^n)^\top$ and

$\theta = (\theta_0, \theta_1, \theta_2, \ldots, \theta_n)^\top$



Model space

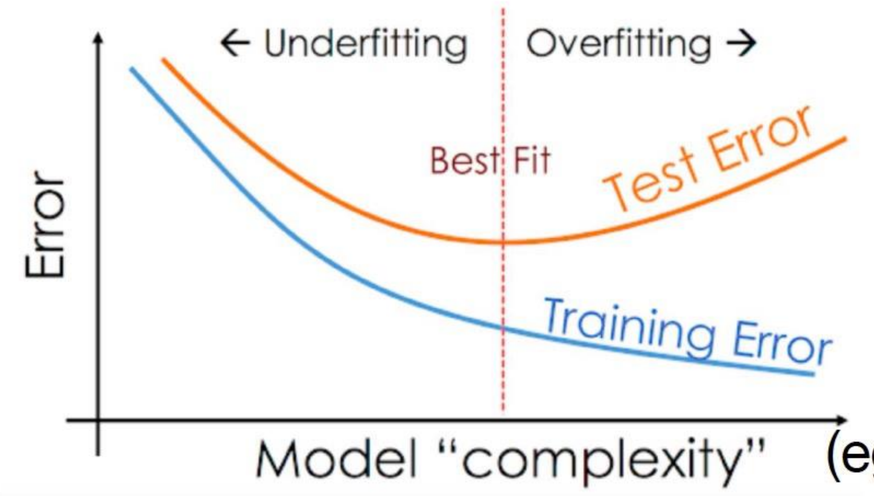
- Which model space should we choose?
- The more complex the model, the large the model space
- Eg. Polynomial function of degree 1, 2, … corresponds to space H1, H2 …



We Need To Select A Proper Model

Intuition of model selection

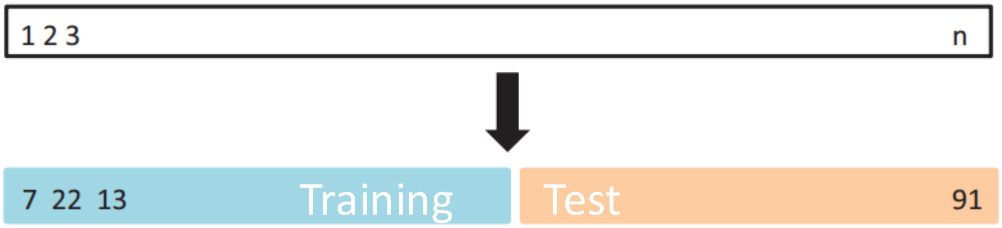
- Find the right model family s.t. test error becomes minimum



(eg, polynomial degree)

Validation Set Approach

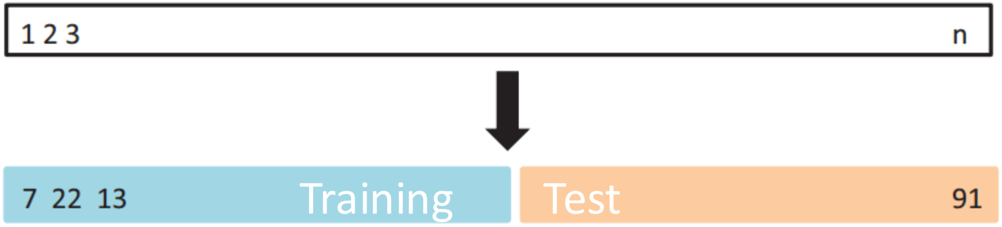
- Divide samples in to **training data** and **test data**.



- A set of model to choose {1,2,…, M}. (e.g, KNN, choose K=1,2,…M)
- For each model m,
  - use training data to train model
  - use the learned model to calculate the prediction error for test data. Err_m
- Choose model that has the smallest test error (min Err_m)

Validation Set Approach

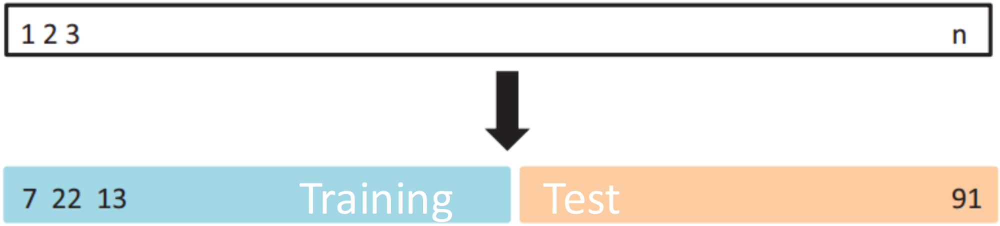
- Divide samples in to training data and test data.



- The selected model highly depends on your separation of the data.
- How to solve?

Validation Set Approach

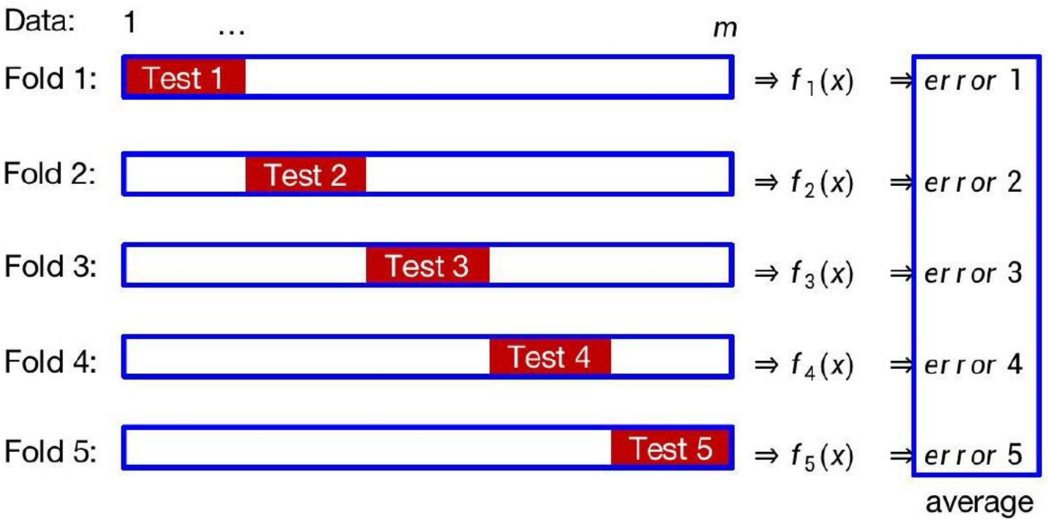
- Divide samples in to training data and test data.



- The selected model highly depends on your separation of the data.
- How to solve?
- **Repeat** the process several time; each time with different training and test data.

K-fold Validation

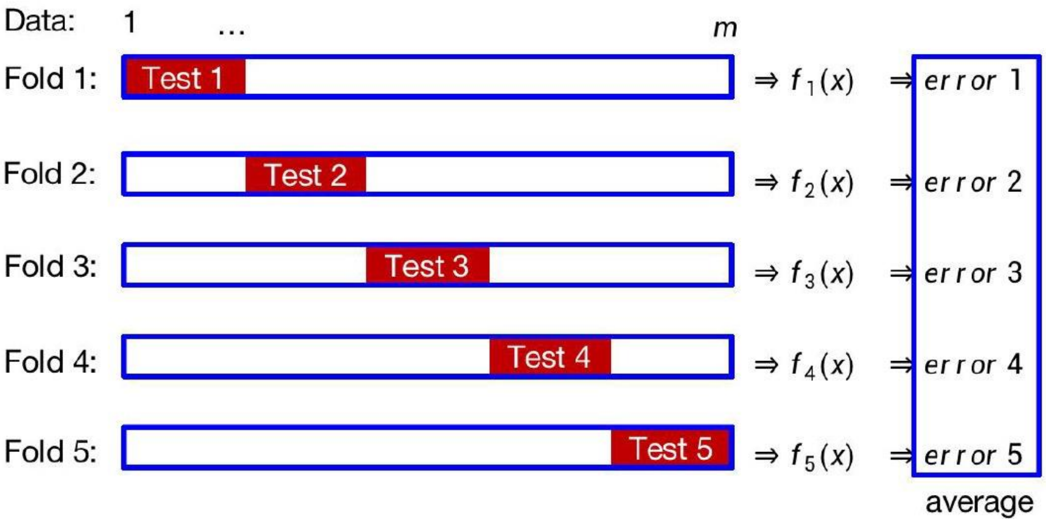
- 5-fold cross-validation (blank: training; red: test)



  - f_i is fitted by the training data in Fold i.
  - For each fold, use test data to test the prediction error.
  - Use the **average** error as the model's prediction error.

K-fold Validation

- 5-fold cross-validation (blank: training; red: test)



Model 1 h(θ, X)	Model 2 g(γ, X)

- f_i is fitted by the training data in Fold i.
- For each fold, use test data to test the prediction error.
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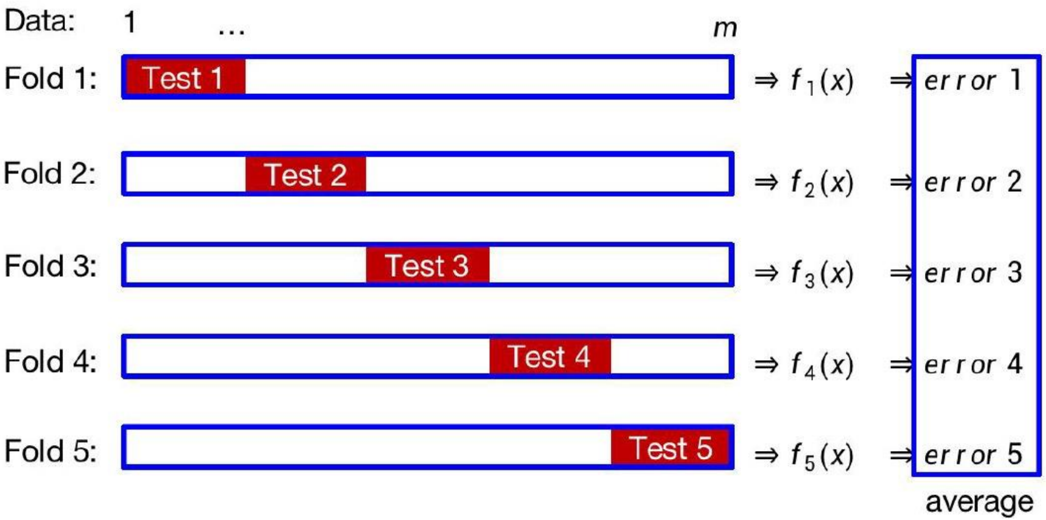
K-fold Validation

- 5-fold cross-validation (blank: training; red: test)



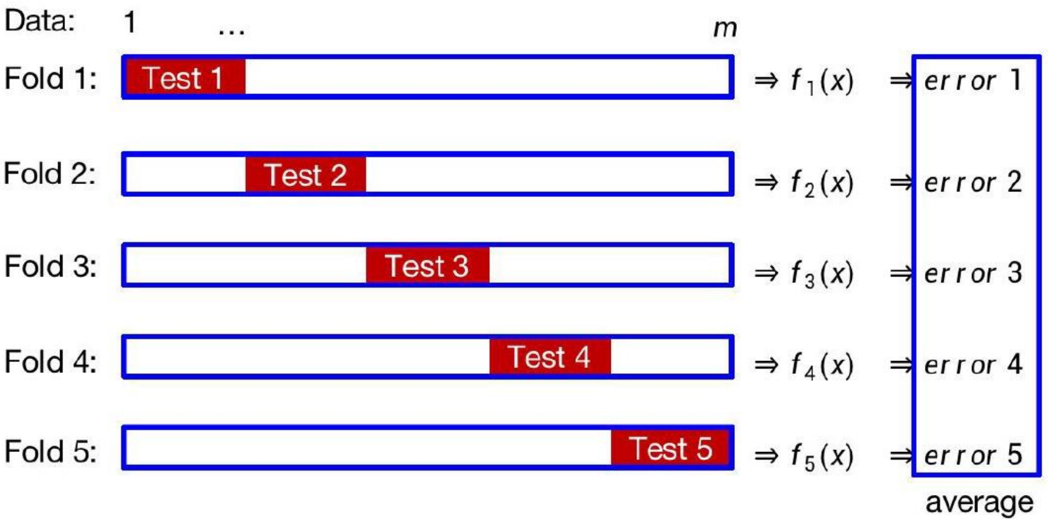
- f_i is fitted by the training data in Fold i.
- For each fold, use test data to test the prediction error.
- Use the **average** error as the model's prediction error.

Use training data to train model

Model 1 $h(\boldsymbol{\theta}, \boldsymbol{X})$	**Model 2** $g(\boldsymbol{\gamma}, \boldsymbol{X})$
$h(\boldsymbol{\theta}_1, \boldsymbol{X})$	$g(\boldsymbol{\gamma}_1, \boldsymbol{X})$
$h(\boldsymbol{\theta}_2, \boldsymbol{X})$	$g(\boldsymbol{\gamma}_2, \boldsymbol{X})$
$h(\boldsymbol{\theta}_3, \boldsymbol{X})$	$g(\boldsymbol{\gamma}_3, \boldsymbol{X})$
$h(\boldsymbol{\theta}_4, \boldsymbol{X})$	$g(\boldsymbol{\gamma}_4, \boldsymbol{X})$
$h(\boldsymbol{\theta}_5, \boldsymbol{X})$	$g(\boldsymbol{\gamma}_5, \boldsymbol{X})$

K-fold Validation

- 5-fold cross-validation (blank: training; red: test)



Error

Model 1 h(θ, X)	**Model 2** g(γ, X)
$\text{Err}_h(\boldsymbol{\theta}_1)$	$\text{Err}_g(\boldsymbol{\gamma}_1)$
$\text{Err}_h(\boldsymbol{\theta}_2)$	$\text{Err}_g(\boldsymbol{\gamma}_2)$
$\text{Err}_h(\boldsymbol{\theta}_3)$	$\text{Err}_g(\boldsymbol{\gamma}_3)$
$\text{Err}_h(\boldsymbol{\theta}_4)$	$\text{Err}_g(\boldsymbol{\gamma}_4)$
$\text{Err}_h(\boldsymbol{\theta}_5)$	$\text{Err}_g(\boldsymbol{\gamma}_5)$

- f_i is fitted by the training data in Fold i.
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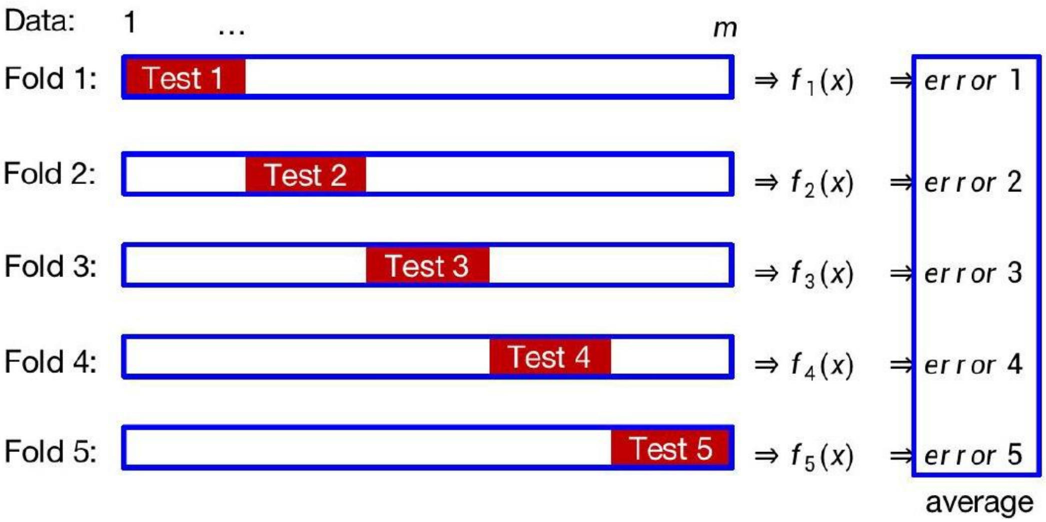
K-fold Validation

- 5-fold cross-validation (blank: training; red: test)



Error

Model 1 $h(\theta, X)$	**Model 2** $g(\gamma, X)$
$\mathrm{Err_h}(\boldsymbol{\theta}_1)$	$\mathrm{Err_g}(\boldsymbol{\gamma}_1)$
$\mathrm{Err_h}(\boldsymbol{\theta}_2)$	$\mathrm{Err_g}(\boldsymbol{\gamma}_2)$
$\mathrm{Err_h}(\boldsymbol{\theta}_3)$	$\mathrm{Err_g}(\boldsymbol{\gamma}_3)$
$\mathrm{Err_h}(\boldsymbol{\theta}_4)$	$\mathrm{Err_g}(\boldsymbol{\gamma}_4)$
$\mathrm{Err_h}(\boldsymbol{\theta}_5)$	$\mathrm{Err_g}(\boldsymbol{\gamma}_5)$

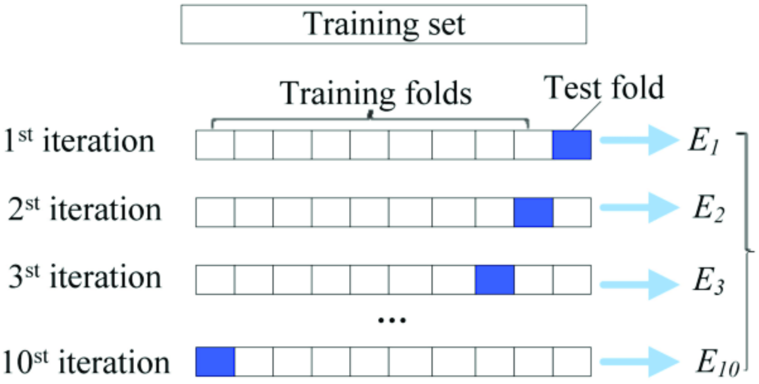
- f_i is fitted by the training data in Fold i.
- For each fold, use test data to test the prediction error.
- Use the **average** error as the model's prediction error.

Model 1 is better iff.

$$\frac{1}{K}\sum_i Err_h(\theta_i) < \frac{1}{K}\sum_i Err_g(\gamma_i)$$

K-fold Validation

- How to decide the values for K (or α)
  - Commonly used $K = 10$ or ($\alpha = 0.1$).
  - Large K makes it time-consuming.



$$E = \frac{1}{K}\sum_{i=1}^{K} E_i$$

Error

Model 1 $h(\theta, X)$	Model 2 $g(\gamma, X)$
$Err_h(\boldsymbol{\theta}_1)$	$Err_g(\boldsymbol{\gamma}_1)$
$Err_h(\boldsymbol{\theta}_2)$	$Err_g(\boldsymbol{\gamma}_2)$
$Err_h(\boldsymbol{\theta}_3)$	$Err_g(\boldsymbol{\gamma}_3)$
$Err_h(\boldsymbol{\theta}_4)$	$Err_g(\boldsymbol{\gamma}_4)$
$Err_h(\boldsymbol{\theta}_5)$	$Err_g(\boldsymbol{\gamma}_5)$

Model 1 is better iff.

$$\frac{1}{K}\sum_i Err_h(\theta_i) < \frac{1}{K}\sum_i Err_g(\gamma_i)$$